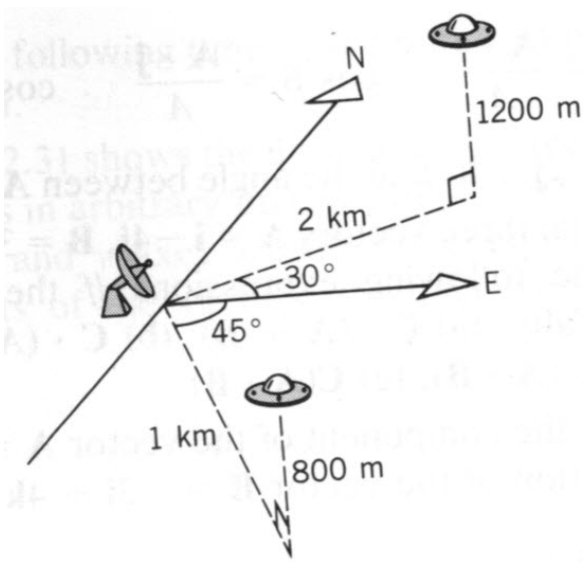
Homework #4

1) Given two vectors, $\vec{A} = 2.00\,\hat{\imath} - 3.00\,\hat{\jmath} + 1.00\,\hat{k}$ and $\vec{B} = -4.00\,\hat{\imath} + 1.00\,\hat{\jmath} - 5.00\,\hat{k}$, find a third vector, $\vec{C}$, such that $\vec{A} - 2.00\vec{B} + \frac{1}{3}\vec{C} = 0$.

2) Given that vector $\vec{A}$ is 5.00 m at 37º North of East, find vector $\vec{B}$ such that their sum is directed East and has a magnitude of 3.00 m. Give the result with respect to compass like how $\vec{A}$ is specified, and with the angle less than or equal to 45º.

3) Personnel at an airport control tower track a UFO. At 11:02 a.m. it is located at a horizontal distance of 2.00 km in the direction 30°N of E at an altitude of 1200 m. At 11:15a.m. the location is 1.00 km at 45° S of E at an altitude of 800 m. What was the displacement of the UFO? Express your result in component notation.



4) The vectors $\vec{A}$ and $\vec{B}$ are in the xy plane where $\vec{A}$ is 3.20 m at 45° to the +x axis, and $\vec{B}$ is 2.40 m at 210^0 to +x axis. (a) What is $\vec{A} \cdot \vec{B}$? (b) What is $\vec{A} \times \vec{B}$?

5) The vectors $\vec{A}$ and $\vec{B}$ are in the xy plane where $\vec{A}$ 3.60 m at 45^0 to the +x axis, and $\vec{B}$ is 4.40 m at 160^0 to the +x axis. (a) What is $\vec{A} \cdot \vec{B}$? (b) What is $\vec{A} \times \vec{B}$?

6) Find a vector of length 5.00 m in the xy plane that is perpendicular to $\vec{A} = 3.00\,\hat{\imath} + 6.00\,\hat{\jmath} - 2.00\,\hat{k}$. (Hint: Consider the dot product.)

7) At $t = 0$ a particle at the origin has a velocity of 15.0 m/s at 37^0 above the horizontal x axis. At $t = 5.00$ s it is at $x = 20.0$ m and $y = 35.0$ m and its velocity is 30.0 m/s at 53^0 above the horizontal. Find: (a) its average velocity; (b) its average acceleration.

8) A plane flies 470 km east from city A to city B in 45.0 min and then 966 km south from city B to city C in 1.50 h. For the total trip, what are the (a) magnitude and (b) direction of the plane's displacement, the (c) magnitude and (d) direction of its average velocity, and (e) its average speed?